$\Rightarrow \quad AS = eAZ \qquad\qquad A'S = eA'Z$

$\Rightarrow \quad CS - CA = e(CA - CZ) \qquad A'C + CS = e(A'C + CZ)$

$\Rightarrow \quad CS - a = e(a - CZ) \ldots (1) \qquad a + CS = e(a + CZ) \ldots (2)$

$(1)+(2)$ gives $CS = ae$ and $(2)-(1)$ gives $CZ = \frac{a}{e}$.

Hence, the coordinates of S are $(ae, 0)$. Since $PM = x - \frac{a}{e}$, the equation of directrix is $x - \frac{a}{e} = 0$.

Let $P(x, y)$ be any point on the hyperbola.

By the definition of a conic, $\frac{SP}{PM} = e \Rightarrow SP^2 = e^2 PM^2$.

Then $(x-ae)^2 + (y-0)^2 = e^2\left(x - \frac{a}{e}\right)^2$

$\Rightarrow \quad (x-ae)^2 + y^2 = (ex - a)^2$

$\Rightarrow \quad (e^2-1)x^2 - y^2 = a^2(e^2-1)$

$\Rightarrow \quad \frac{x^2}{a^2} - \frac{y^2}{a^2(e^2-1)} = 1$. Since $e > 1, a^2\left(e^2-1\right) > 0$. Setting $a^2(e^2-1) = b^2$, we obtain the locus of P as

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which is the equation of a **Hyperbola in standard form** and note that it is symmetrical about x and y-axes.

Taking $ae = c$, we get $b^2 = c^2 - a^2$.

Definition 5.5

(1) The line segment AA' is the **transverse** axis of length $2a$.

(2) The line segment BB' is the **conjugate** axis of length $2b$.

(3) The line segment $CA =$ the line segment $CA' =$ **semi transverse axis** $= a$ and the line segment $CB =$ the line segment $CB' =$ **semi conjugate axis** $= b$.

(4) By symmetry, taking $S'(-ae, 0)$ as focus and $x = -\frac{a}{e}$ as directrix l' gives the same hyperbola.

Thus we see that a hyperbola has two foci $S(ae, 0)$ and $S'(-ae, 0)$, two vertices $A(a, 0)$ and $A'(-a, 0)$ and two directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$.

Length of latus rectum of hyperbola is $\frac{2b^2}{a}$, which can be obtained along lines as that of the ellipse.

Asymptotes

Let $P(x, y)$ be a point on the curve defined by $y = f(x)$, which moves further and further away from the origin such that the distance between P and some fixed line tends to zero. This fixed line is called an **asymptote**.

Note that the hyperbolas admit asymptotes while parabolas and ellipses do not.

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